Sign of permutation.

Let a polynomial over $\{\pm 1\}$ with distinct $n$-variables $x_1, \dots, x_n$ as

$$\Delta \overset{\text{def}}{=} \prod_{1 \le i < j \le n} (x_i - x_j)$$

For each $\sigma \in S_n$, let $\sigma$ act on $\Delta$ by permuting $i$

$$\sigma(\Delta) = \prod_{1 \le i < j \le n} (x_{\sigma(i)} - x_{\sigma(j)})$$

Since $\sigma$ is bijective, there are two possibities;

$$\sigma(\Delta) = \pm \Delta.$$

Now, let $\operatorname{sign}: S_n \to \{\pm 1\} : \sigma \mapsto \begin{cases} 1 & \text{if } \sigma(\Delta) = \Delta \\ -1 & \text{if } \sigma(\Delta) = -\Delta. \end{cases}$

Prop. Sign is a surjective homomorphism.

Proof. Given $\sigma, \tau \in S_n$,

$$(\tau \circ \sigma)(\Delta) = \prod_{1 \le i < j \le n} (x_{(\tau\circ\sigma)(i)} - x_{(\tau\circ\sigma)(j)})$$

Suppose $\sigma(\Delta)$ has exactly $k$ factors of $(x_j - x_i)$ with $j > i$. That is, $\operatorname{sign} \sigma = (-1)^k$.

So, $$\sigma(\Delta) = \prod_{1 \le i < j \le n} (x_{\sigma(i)} - x_{\sigma(j)}) = \operatorname{sign}\sigma \cdot \prod_{1 \le p < q \le n} (x_p - x_q)$$

Therefore, $(\tau \circ \sigma)(\Delta) = \operatorname{sign}\sigma \cdot \prod_{1 \le p < q \le n} (x_{\tau(p)} - x_{\tau(q)})$, it follows

$$\operatorname{sign}(\tau \circ \sigma) = \operatorname{sign}\sigma \cdot \operatorname{sign}\tau.$$

Surjectiveness; clearly, $\operatorname{sign}(id) = 1$. And, $\operatorname{sign}((1,2)) = -1$, so clear. ∎

Prop. $\ker \operatorname{sign} = A_n$.

Prop. $m$-cycle is odd iff $m$ is even

Prop. a permutation $\sigma$ is odd iff number of cycles of even length is odd, in decomposition.

Prop. $S_n$ is generated by $\{(i\ \ i+1) \mid 1 \le i \le n-1\}$.

Proof. Given $j \in \{2, \cdots, n-1\}$,

$$(1\ 2\ 3\ \cdots\ n) = (1\ n)(1\ n-1)\cdots(1\ 3)(1\ 2)$$

$$(1\ 2)(2\ 3)\cdots(j-1\ \ j) = (1\ 2\ 3\ \cdots\ j)$$

So, the generated set contains $(1\ 2\ \cdots\ j)$ for all $2 \le j \le n$.

Moreover, observe that

$$(2\ 3)(1\ 2)(2\ 3)^{-1} = (2\ 3)(1\ 2\ 3) = (1\ 3)$$

$$(3\ 4)(1\ 3)(3\ 4) = (3\ 4)(1\ 3\ 4) = (1\ 4)$$

$$\vdots$$

$$(j\ j+1)(1\ j)(j\ j+1) = (1\ \ j)$$

Now, for any $1 \le j < k \le n$,

$$(j\ \ k) = (1\ k)(1\ j)$$

can be generated by the given set. All permutations can be represented by product of transpositions, so $S_n = \langle (i\ \ i+1) \mid 1 \le i \le n-1 \rangle$

Corollary. $S_n = \langle (1\ 2), (1\ 2\ \cdots\ n) \rangle$

Proof. Observe that

$$(1\ 2\ \cdots\ n)(1\ 2)(1\ 2\ \cdots\ n)^{-1} = (1\ 2\ \cdots\ n)(1\ 2)(n\ n-1\ \cdots\ 1) = (2\ 3)$$

$$(1\ 2\ \cdots\ n)(2\ 3)(1\ 2\ \cdots\ n)^{-1} = (3\ 4)$$

$$\vdots$$

So, $\langle (1,2), (1\ 2\ \cdots\ n) \rangle$ contains $(i\ \ i+1)$ for all $1 \le i \le n-1$, proved

+ Note that

$$\begin{aligned}
&(a_1\ a_2\ \cdots\ a_n) \\
&= (a_1\ a_n)(a_1\ a_2\ \cdots\ a_{n-1}) \\
&= (a_1\ a_n)(a_1\ a_{n-1})(a_1\ a_2\ \cdots\ a_{n-2}) \\
&\ \ \vdots \\
&= (a_1\ a_n)(a_1\ a_{n-1})\cdots(a_1\ a_2)
\end{aligned}$$

[Prop. For a prime number $p$, $S_p$ is generated by any one transposition and $p$-cycle.

Pf. WLOG, let the one transposition is $(1\ x)$ and $p$-cycle is $(1\ \cdots\ p)$. where $x \in \{2, \cdots, p\}$. And, since $p$ is prime number,

$$(1\ 2\ \cdots\ p)^k \quad (1 \le k \le p-1)$$

are all $p$-cycles. And, take $k = x-1$, then

$$(1\ 2\ \cdots\ p)^k = (1\ x\ \cdots)$$

So $\langle (1\ x), (1\ \cdots\ p) \rangle = \langle (1\ x), (1\ x\ \cdots) \rangle = S_p$.

Note; $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$ cannot generate $S_4$. Observe that

$(1\ 3)^2 = id$, $(1\ 2\ 3\ 4)^2 = (1\ 3)(2\ 4)$, $(1\ 2\ 3\ 4)^3 = (1\ 4\ 3\ 2)$

$(1\ 3)(1\ 2\ 3\ 4) = (1\ 2)(3\ 4)$

$(1\ 3)(1\ 2\ 3\ 4)^2 = (2\ 4)$

$(1\ 3)(1\ 2\ 3\ 4)^3 = (1\ 4)(2\ 3)$

So, $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle = \langle (1\ 2)(3\ 4), (1\ 3)(2\ 4) \rangle$.

[Prop. $A_4$ has No subgroup of order 6.

Proof. Let $V = \{id, (12)(34), (13)(24), (14)(23)\}$ be a subset of $A_4$.
Except $id \in V$, the elements of $V$ have order of 2, exactly,
these are all elements of order 2 in $A_4$.
The Conjugate cannot change the order and parity, $V$ is a Normal subgroup of $A_4$
And, since $A_4/V$ has order of 3, it must be contain order 3 element,
so, it must be isomorphic to $\mathbb{Z}_3$, cyclic.
Meanwhile, if there exists a order of 6 subgroup, $N$, then
since $[A_4 : N] = 2$, thus for any $g \in A_4$, $gN = Ng$. so, it is normal.
Now, $A_4/N \cong \mathbb{Z}_2$, cyclic.
Meanwhile, observe that; for $G' = \{x^{-1}y^{-1}xy \mid x, y \in G\}$, and $N \trianglelefteq G$,

$G/N$ is abelian iff $G' \leq N$

To show this, Let $x^{-1}y^{-1}xy \in G'$ be given. Since $G/N$ is abelian,

$$x^{-1}y^{-1}xyN = x^{-1}N\,y^{-1}N\,xN\,yN = x^{-1}Nx\,Ny^{-1}Ny\,N = N$$

so $x^{-1}y^{-1}xy \in N$.
Therefore, $G' \leq N$ and $G' \leq V$. And, since $A_4$ is Not abelian, $G' \neq 1$.
And, if $|G'| = 2$, then it must be contain an order of 2 element,
but since $G'$ is normal, then it has all order of 2 element.
Thus, $G' \geq V$, or $G' = V$. Now, $|G'| = 4$ but $G' \not\leq N$,
it is contradiction to the Lagrange Thm.